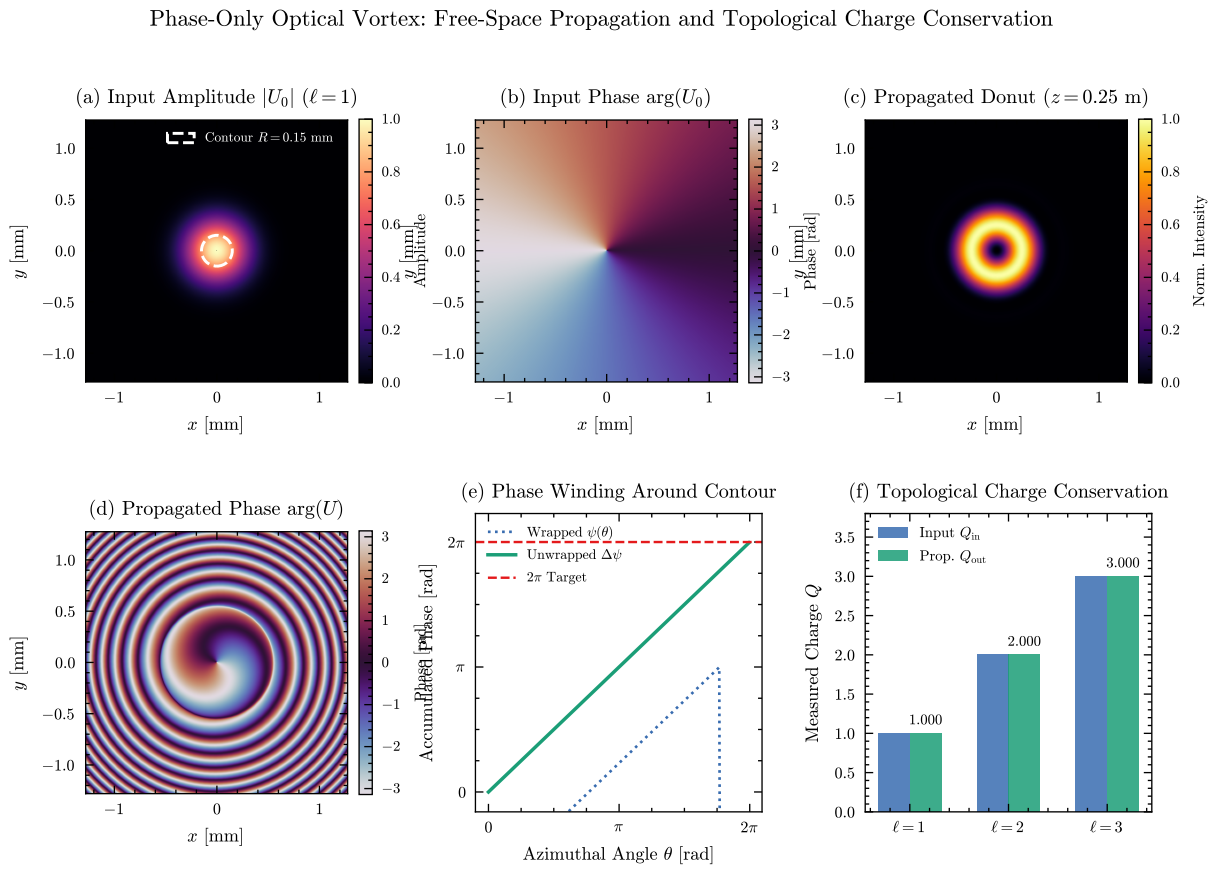

Phase-Only Optical Vortex: Free-Space Propagation and Topological Charge Conservation
(a) Input Amplitude $|U_0|$ ($\ell = 1$)
Contour $R = 0.15$ mm
(b) Input Phase arg(U_0)
(c) Propagated Donut ($z = 0.25$ m)
Norm. Intensity
(d) Propagated Phase arg(U)
(e) Phase Winding Around Contour
Wrapped $\psi(\theta)$
Unwrapped $\Delta\psi$
2π Target
Accumulated Phase [rad]
Azimuthal Angle θ [rad]
(f) Topological Charge Conservation
Input Q_{in}
Prop. Q_{out}
Measured Charge Q
1.000
2.000
3.000
$\ell = 1$
$\ell = 2$
$\ell = 3$
x [mm]
y [mm]